Topic 11 — Ozone Layer

Covers syllabus (click to expand)

Ozone Layer | Ozone Depletion (ओजोन क्षरण) | Ozone Hole (ओजोन छिद्र) | Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल)

Sources baked in: NCERT Geography/Biology, UNEP Ozone Secretariat, Vienna Convention 1985, Montreal Protocol 1987, WMO assessments, UPPCS PYQs 2018–2025 **Weight:** ★★ Medium–High — CFC mechanism, treaty traps, ozone hole geography **Last verified:** August 2026

Current Affairs (this topic)

- Montreal / Kigali compliance headlines stay ozone-family CA — do not park them under Paris NDC.
- ODS vs GHG trap is permanent; HFC phase-down is Kigali, not Kyoto (क्योटो).

11.1 Ozone Layer

Identity: O_3 in stratosphere | UV shield | Chapman cycle | Dobson Unit

- **Ozone (O_3)** is a molecule of three oxygen atoms that absorbs harmful ultraviolet radiation.
- The **ozone layer** lies mainly in the **stratosphere** (समतापमंडल) at about **15–35 km**, with peak concentration near **25 km**.
- The **Chapman cycle** describes natural ozone formation when UV splits O_2 and O combines with O_2 to form O_3.
- Ozone absorbs UV energy and releases heat, which is why **stratosphere temperature rises with height**.
- **UV-A** largely reaches the surface. **UV-B** is mostly blocked by ozone and causes sunburn and skin cancer if exposure rises.
- **UV-C** is fully absorbed by ozone and oxygen under normal conditions.
- **Column ozone** is measured in **Dobson Units (DU)**. Global average is about **300 DU**.
- **Stratospheric ozone** is protective. **Tropospheric ozone** is a harmful air pollutant and greenhouse gas.
- Without the ozone layer, lethal UV levels would reach Earth's surface and damage DNA, crops, and marine plankton.

Logic: UPPCS 2023 pattern — ozone layer is in the **stratosphere**, not troposphere.

PYQ — UPPCS Prelims 2025, Q45

1. The ozone layer protects Earth from ultraviolet radiation.

2. Ozone depletion has been linked to CFCs.

A. Only 2

B. Neither

C. Both 1 and 2

D. Only 1

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Ans: C — Both statements are correct.

UV type	Wavelength	Effect if unfiltered
UVA	320–400 nm	Ageing, tanning
UVB	280–320 nm	Sunburn, skin cancer, cataracts
UVC	100–280 nm	Lethal — normally blocked

11.2 Ozone Depletion

Identity: CFCs/ODS | Catalytic chlorine | Rowland–Molina 1974

- **Ozone depletion** is the sustained reduction of stratospheric ozone below natural levels.
- **Ozone-Depleting Substances (ODS)** contain chlorine and/or bromine that destroy ozone in the stratosphere.
- **CFCs** were used in refrigerants, aerosols, foam blowing, and solvents.
- CFCs are stable in the troposphere and take decades to reach the stratosphere.
- UV breaks CFC bonds and releases **chlorine atoms (Cl·)** that destroy ozone in a **catalytic cycle**.
- **One chlorine atom can destroy up to about 100,000 ozone molecules** before removal.
- **Halons** used in fire suppressants release **bromine**, which is even more destructive per atom.
- Other ODS include **carbon tetrachloride**, **methyl bromide**, and transitional **HCFCs**.
- The **Rowland–Molina hypothesis (1974)** linked CFCs to ozone loss and won the Nobel Prize in 1995.
- Increased UV-B raises **skin cancer, cataracts, crop damage, and marine food chain (खाद्य श्रृंखला) stress**.
- After the Montreal Protocol, ODS emissions fell about **99%**. Mid-latitude recovery is expected around **2040**.

> **Logic:** CFCs destroy ozone by **UV chemistry**, not by greenhouse trapping — that is a separate effect.

ODS	Main use
CFCs	Refrigerants, aerosols, foam

ODS	Main use
Halons	Fire extinguishers
Carbon tetrachloride	Solvent
Methyl bromide	Pesticide/fumigant
HCFCs	Transitional refrigerants

11.3 Ozone Hole

Identity: < 220 DU | Antarctic spring | PSCs | Not a physical hole

- The **ozone hole** is **not a hole in the atmosphere**. It is a region where ozone falls below **220 DU**.
- The most severe and recurring depletion occurs over **Antarctica** in **Southern Hemisphere spring (September–October)**.
- The **polar vortex** isolates cold Antarctic stratospheric air during winter darkness.
- **Polar Stratospheric Clouds (PSCs)** (ध्रुवीय समतापमंडलीय बादल) form at very low temperatures and enable chlorine activation.
- Spring sunlight triggers rapid ozone destruction when the polar vortex breaks down.
- The ozone hole was discovered in **1985** by **Farman and colleagues** at the British Antarctic Survey.
- **Arctic** ozone loss can occur in cold winters but is **less severe and less regular** than Antarctic depletion.
- India is not under the ozone hole, but **Himalayan high altitude** increases UV-B exposure risk.
- WMO projects **Antarctic recovery to 1980 levels around 2066** if Montreal controls continue.

> **Logic:** Trap — "ozone hole over Arctic every winter" is **false**. Classic hole = **Antarctic spring**.

Feature	Antarctic	Arctic
Severity	Severe, recurring	Occasional, milder
Season	Sep–Oct (austral spring)	Mar–Apr if event occurs
Polar vortex	Strong, very cold	Weaker, warmer

11.4 Montreal Protocol

Identity: Vienna 1985 + Montreal 1987 | ODS phase-out | Kigali HFCs | India compliance

- The **Vienna Convention (1985)** (वियना कन्वेंशन) created the framework to protect the ozone layer through cooperation and monitoring.
- The **Montreal Protocol (1987)** (मॉन्ट्रियल प्रोटोकॉल) set **binding schedules** to phase out production and consumption of ODS.

- **International Ozone Day** is **16 September**, the date the Montreal Protocol was signed.
- The **Multilateral Fund (1991)** helps developing countries switch to ODS-free technologies.
- The Montreal Protocol is the **only UN treaty ratified by every country**.
- ODS emissions have declined about **99%** since their peak in the 1980s–1990s.
- **India ratified in 1992**. The **Ozone Cell under MoEFCC** coordinates compliance.
- India ended **CFC production in 2010** and targets **HCFC phase-out by 2030**.
- The **Kigali Amendment (2016)** (किगाली संशोधन) phases down **HFCs**, which do not deplete ozone but have **high global warming (भूमंडलीय तापन) potential**.
- **Montreal controls ozone**. **Kyoto controls climate**. Do not swap them.

PYQ — UPPCS Prelims 2024, Q35

Which agreement relates to climate change (जलवायु परिवर्तन) mitigation?

A. Montreal Protocol

B. Ramsar (रामसर)

C. Kyoto Protocol

D. CITES (साइट्स)

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Ans: C — Montreal = **ozone/ODS**.

Treaty / step	Year	Role
Vienna Convention	1985	Framework
Montreal Protocol	1987	Binding ODS phase-out
London Amendment	1990	Multilateral Fund
Kigali Amendment	2016	HFC phase-down